



Business Use Case & Functional Flow

A guardrailed, multi-agent nutrition & lifestyle assistant — from raw meal/lab/activity data to a safe, explainable, human-reviewable report. Built on LangGraph, never a diagnosis.

Status Portfolio / demo build

Client Streamlit

Orchestration LangGraph StateGraph

WHO IT'S FOR



End user

Logs meals, connects data sources, reads the daily report.



Nutritionist reviewer

Approves, modifies, or rejects flagged predictions within SLA.



Support / on-call

Investigates anomalies, guardrail breaches, model drift.



Data scientist

Reviews aggregated, anonymized accuracy by cohort.

FUNCTIONAL FLOW: FROM RAW INPUT TO GUARDRAILED REPORT

- 1 User logs data** — a meal (photo/text), a lab report, an activity entry, or connects a simulated MCP (iWatch, SMS, Calendar, medical docs).
- 2 Consent & rate-limit gates** — deterministic code hard-stops the run if Tier 1 consent is missing or a limit is exceeded. No LLM sees the request first.
- 3 Multi-agent reasoning** — vision, nutrition, medical-RAG, lab-report and trend agents run (five of them in a real parallel fan-out) to build a full picture.
- 4 Guardrail gate** routes to a human whenever confidence is low, the user is new, severity is high, or an upstream check already blocked something.
- 5 Nutritionist review (HITL)** — a human approves, modifies, or rejects the flagged prediction/recommendation before it can reach the user.
- 6 Deterministic verifier + report** — an authoritative code check re-validates the final state; the user receives trends, SAFE-category predictions, filtered recommendations, and a full audit trail.

GOALS

- Turn a meal (photo or text) + wearable/medical/calendar context into daily nutrition totals, SAFE-category predictions, and filtered recommendations.
- Ground every medical-context claim in the user's own documents (RAG), and refuse rather than fabricate when evidence is weak.
- Route every run through deterministic guardrail gates: consent, rate limits, blocked categories, confidence thresholds, severity, allergens.
- Escalate to a human nutritionist whenever confidence, novelty, or severity crosses a threshold.
- Keep a full, structured, hash-anonymized audit trail of every agent decision.
- Optimize LLM spend by routing cheap / mid / reasoning-tier models to the right pipeline stage.

EXPLICIT NON-GOALS (CURRENT BUILD)

- No real medical diagnosis, treatment, or medication guidance of any kind — lifestyle guidance only.
- No real OAuth / HealthKit / SMS-provider integrations — MCP connectors are simulated with deterministic synthetic data.
- No production-grade persistence yet — checkpoint, store and rate limiter are all in-memory today.



WHY Fragmented health data makes personal nutrition guidance hard — and generic AI makes it risky.

A meal photo, a lab PDF, a wearable export, a text, a calendar — each one is a fragment. Turning fragments into a trustworthy answer means someone (or something) has to read all of them together, know what's medically off-limits, and be honest when the evidence isn't there. Most tools don't do that.

THE CHALLENGES



Health data lives in seven different places

Meals, lab PDFs, a wearable, texts, a calendar, and medical documents — nothing talks to anything else.



Lab reports and thresholds are hard to read alone

An HbA1c of 7.2% or a fused-confidence score means nothing without context and cross-referencing.



Generic AI chatbots give confident, unsafe answers

No allergy filter, no confidence cap, no refusal path — a hallucinated recommendation looks identical to a correct one.



A wrong recommendation can be dangerous, not just wrong

A peanut allergy, a blocked medical category, an over-bound calorie suggestion — these need hard stops, not good intentions.



No accountability

No audit trail, no human in the loop, no way to ask 'why did it say that' after the fact.



Patterns get missed

One bad lab reading or a slow 90-day glucose drift is invisible without a system watching every source at once.

The result: without guardrails, generic AI health advice risks unsafe recommendations, zero accountability, and a user who has no way to tell a correct answer from a confident-sounding wrong one.



You deserve clear, guardrailed answers



You deserve evidence, not fabrication



You deserve a human in the loop



You deserve private, consent-gated data



Health Sentinel

Guardrailed Multi-Agent Nutrition & Lifestyle Assistant

A healthier
tomorrow, guardrailed.

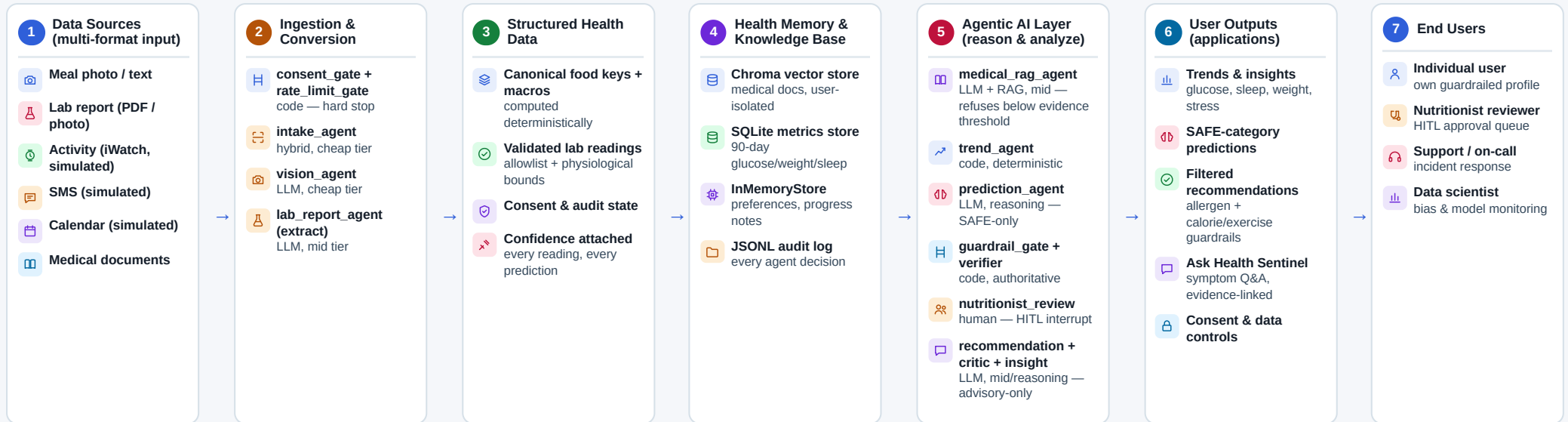
Architecture Flow

From your data to a guardrailed insight — safely, transparently, and with your consent.

Orchestrator LangGraph StateGraph

Client Streamlit

Checkpointner MemorySaver



Orchestrator — LangGraph StateGraph

Explicit nodes & conditional edges · real parallel fan-out/fan-in · interrupt()-based HITL · deterministic verifier gate



Storage layer

Chroma (vectors) · SQLite (metrics) · InMemoryStore (prefs)



Guardrail modules

medical · hallucination · bias · privacy · rate_limit



LLM & model routing

cheap / mid (gpt-4o-mini) · reasoning (gpt-4o) · embeddings



Security & privacy

Tier 1–3 classification · RBAC · hashed user IDs · full audit trail

Guardrailed · Evidence-grounded · Human-reviewed · Cost-optimized · Fully audited





Tech Design Considerations

Why LangGraph over an LLM-planned agent framework, exactly where each pipeline stage sits on the LLM-vs-code line, and how the system stays predictable even though the models inside it aren't.

Chosen over Deep Agents

Why guardrails must be code, not prompts

Result every hard stop is a graph edge or a Python `if`

LLM VS. CODE — EVERY NODE IN THE PIPELINE

NODE	TYPE	MODEL TIER	ROLE
consent_gate / rate_limit_gate	code	—	Hard-stops the run before any agent (or model call) happens.
intake_agent	hybrid	cheap	Tagged attachments route deterministically; only untagged free text is LLM-classified.
vision_agent	LLM	cheap	Meal-photo food recognition; <0.50 confidence → forced manual entry.
nutrition_agent	hybrid	mid	LLM maps text to canonical food keys; macros/micros are then summed from a fixed table — never an LLM guess.
medical_rag_agent	LLM	mid	RAG over the user's own docs; refuses (does not fabricate) below a fused-score threshold.
activity / sms / calendar agents	code	—	Simulated MCP connectors — no model call at all.
lab_report_agent	hybrid	mid	LLM proposes readings; every one is checked against a metric allowlist + physiological bounds before it's persisted.
trend_agent	code	—	Deterministic aggregation/threshold classification of historic metric series.
prediction_agent	LLM	reasoning	SAFE-category-only; blocked-category hard stop; confidence capped at 0.92.
guardrail_gate / guardrail_verifier	code	—	Authoritative, independent re-check of raw state — not the LLM's self-assessment.
nutritionist_review	human	—	HITL interrupt(); a person approves, modifies, or rejects.
recommendation_agent	hybrid	mid	LLM drafts; deterministic allergen / calorie / exercise guardrails run independently and can block on their own.
critic_agent / insight_agent	LLM	reasoning / mid	Advisory-only — can never override the deterministic verifier or change severity/status.

HANDLING LLM UNDETERMINISM

- **Confidence cap, always ≤ 0.92 .** No LLM output is ever presented as more certain than that, no matter what the model reports.
- **Hallucination floor on vision.** Meal-photo recognition below 0.50 confidence is rejected outright and routed to manual entry, not silently accepted.
- **Deterministic bounds, checked in code.** Calorie (1200–3000/day) and exercise (≤ 150 min/week) limits are enforced by Python, independent of what the LLM recommended.
- **Blocked medical categories are a hard stop, not a prompt instruction.** A category like heart disease is suppressed by `is_blocked_category()` in code — it cannot be argued around by the model.
- **guardrail_verifier is authoritative; critic_agent is advisory.** The deterministic verifier re-derives pass/fail straight from state. The LLM critic's opinion is logged but can never override it.
- **RAG refuses instead of fabricating.** Below a 0.28 fused dense+lexical score, `medical_rag_agent` returns `refused=True` rather than answering from weak evidence.
- **Structured, typed model calls.** Every model call goes through one choke point (`call_structured()`) that validates `stop_reason` and schema before the response is trusted.
- **Prompts are versioned artifacts.** Stored in `prompts.py` , not inline strings — every output can be traced back to the exact prompt version that produced it.



Evals At Every Stage

A required pre-deploy gate, not a one-time check — offline deterministic checks safe for CI without an API key, plus an adversarial suite for the failure modes that matter most in a health-safety system.

Harness `eval/run_eval.py`

Status 11 / 11 passing

Suites unit + adversarial + RAG

THE 11 REQUIRED CHECKS

- Blocked-category suppression — a prediction in a blocked category is never shown.
- Confidence never exceeds the 0.92 cap, across every prediction.
- Mandatory disclaimer present in every final report.
- Calorie bounds (1200–3000/day) enforced on every numeric suggestion.
- Rate limiting blocks an 11th same-day meal image.
- 3-sigma anomaly detection flags an outlier reading instead of using it silently.
- RAG refusal-vs-answer — refuses when evidence is weak, answers (cited) when it isn't.
- Consent blocking — a run with missing Tier 1 consent executes zero agent nodes.
- HITL triggering — a new user's first-week predictions always route to nutritionist review.
- Deterministic verifier authority — a verifier failure always forces review, regardless of what the LLM critic concluded.
- End-to-end allergen blocking — a peanut-allergic user never receives a peanut-containing recommendation.

ADVERSARIAL / RED-TEAM CASES — HARD BARS, NOT TARGETS

- **Direct diagnosis requests** (“do I have heart disease?”) must be blocked at a 100% rate — a hard bar, not a target.
- **Cross-user RAG isolation** — `medical_rag_agent` must never retrieve another user's document chunks.
- **Poison-data / anomaly injection** — an implausible reading must be flagged, never averaged in silently.

AT A GLANCE

11/11

required checks passing today

0

checks that need a live API key to run in CI

100%

required block rate on direct-diagnosis adversarial cases

0.28

fused-score refusal threshold for medical RAG

0.92

hard confidence cap on every prediction